

The empirical recurrence for OEIS A300802, recovered from its tail

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8 September 2026

Abstract

OEIS A300802 records “Empirical recurrence of order 85” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 190$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 88$. Its last coefficient is $c_{85} = 8230118400 \neq 0$, so no further tail satisfies anything shorter; any order-85 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A300802 is “Number of $n \times 6$ 0..1 arrays with every element equal to 0, 1, 2, 3 or 4 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

32, 2048, 89472, 4369224, 210678360, 10189867088, 492720806016, 23826737802264, ...

The entry states:

Empirical recurrence of order 85 (see link above)

As of the “Last modified” line on the live entry (Mar 13 2018 07:44:21, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 190$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 190$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 85: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 85.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 380$ exact terms from $n = 3$ on, it returns order 85 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{85} c_i a(n-i),$$

c_1	45	c_{23}	522323631668280	c_{45}	285361551860577422822	c_{67}	−132101831266715
c_2	246	c_{24}	−951879022115020	c_{46}	−462056915292789856537	c_{68}	−262118374075024
c_3	−3575	c_{25}	4377519259746905	c_{47}	449567234265163008210	c_{69}	−294383769981258
c_4	−25070	c_{26}	−1700337760183489	c_{48}	−571476638177035766304	c_{70}	1426188597217760
c_5	95051	c_{27}	20317410495427752	c_{49}	686656825269122946128	c_{71}	−50399044080853
c_6	878863	c_{28}	407928309326696	c_{50}	−289733138065396866460	c_{72}	192881814674243
c_7	−163549	c_{29}	130728043408732306	c_{51}	−195022704463506373533	c_{73}	−15404403373937
c_8	−9578007	c_{30}	34966833833015028	c_{52}	462639024478644890861	c_{74}	178184138976468
c_9	−16936879	c_{31}	303901431817191410	c_{53}	−291451844055606388165	c_{75}	97729854796047
c_{10}	−95978077	c_{32}	−267170572450615350	c_{54}	−181790433092198784763	c_{76}	−7278376245209
c_{11}	−68716203	c_{33}	393613830752152120	c_{55}	222065051255796993637	c_{77}	−4304049883470
c_{12}	1023979879	c_{34}	−1942484331043237581	c_{56}	−14667047342335118835	c_{78}	−4706020919106
c_{13}	75017506	c_{35}	−1124070715307864850	c_{57}	−45006557171491204432	c_{79}	3287660914728
c_{14}	−40419126512	c_{36}	−7273530902384381399	c_{58}	−31613325794457760982	c_{80}	−1453279543236
c_{15}	−20578568089	c_{37}	1446643092803843342	c_{59}	52371688059709496497	c_{81}	311469559929
c_{16}	−321656963925	c_{38}	−10051000617871455616	c_{60}	−3434836622954076785	c_{82}	−269647779492
c_{17}	1322291141548	c_{39}	18702966791116213025	c_{61}	7232656019517846016	c_{83}	116377093440
c_{18}	−616587058424	c_{40}	−13823046791600635407	c_{62}	−7566645986700271915	c_{84}	2434323456
c_{19}	13552686066532	c_{41}	84254926907454010089	c_{63}	−1742822272570261456	c_{85}	8230118400
c_{20}	7059198156351	c_{42}	−55692910310416138410	c_{64}	−513839475648218091		
c_{21}	184138835958556	c_{43}	137723530716935926575	c_{65}	1217569617426670094		
c_{22}	521511605223	c_{44}	−237476764961763638207	c_{66}	156245428822438090		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 88$, and it is the recurrence OEIS A300802 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{85} - c_1x^{84} - \dots - c_{85}$. Its constant term is $-c_{85} = -8230118400$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 85 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 85, so $\deg q = 85$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 85, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 85, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 8230118400.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-85 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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